

IAM Governance Foundations

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Building a Strong Foundation for Identity Management

Effective IAM governance is the cornerstone of enterprise security. This notebook establishes the framework for managing identities, groups, policies, and access controls in Dynatrace's Gen3 IAM system.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM enabled
Permissions	Account Administrator or IAM Administrator
Knowledge	Basic understanding of identity concepts

1. Gen3 IAM Overview

Dynatrace Gen3 IAM replaces the legacy Management Zone-based access control with a modern **Attribute-Based Access Control (ABAC)** model. This provides fine-grained, scalable access management for enterprises.

Key Components

Component	Purpose	Scope
Policies	Define what actions users can perform	Account or Environment
Boundaries	Filter which entities users can see	Environment
Buckets	Physically partition data for team isolation	Environment
Segments	Reusable DQL-based data filters	Environment
Groups	Collections of users with shared access	Account
Users	Individual identities (local or federated)	Account

Data Partitioning Options

Choose the right isolation mechanism for your needs:

Mechanism	Isolation Type	Reversible	Best For
Buckets	Physical separation	✗ No	Strict team isolation, compliance
Boundaries	Query-time filtering	✓ Yes	Flexible access scoping
Segments	Query-time filtering	✓ Yes	Ad-hoc data views

⚠ **Key Decision:** If you need **hard data isolation** between teams (e.g., Team A can never see Team B's data), use **Buckets**. If you need flexible, changeable access controls, use **Boundaries**.

Why Gen3?

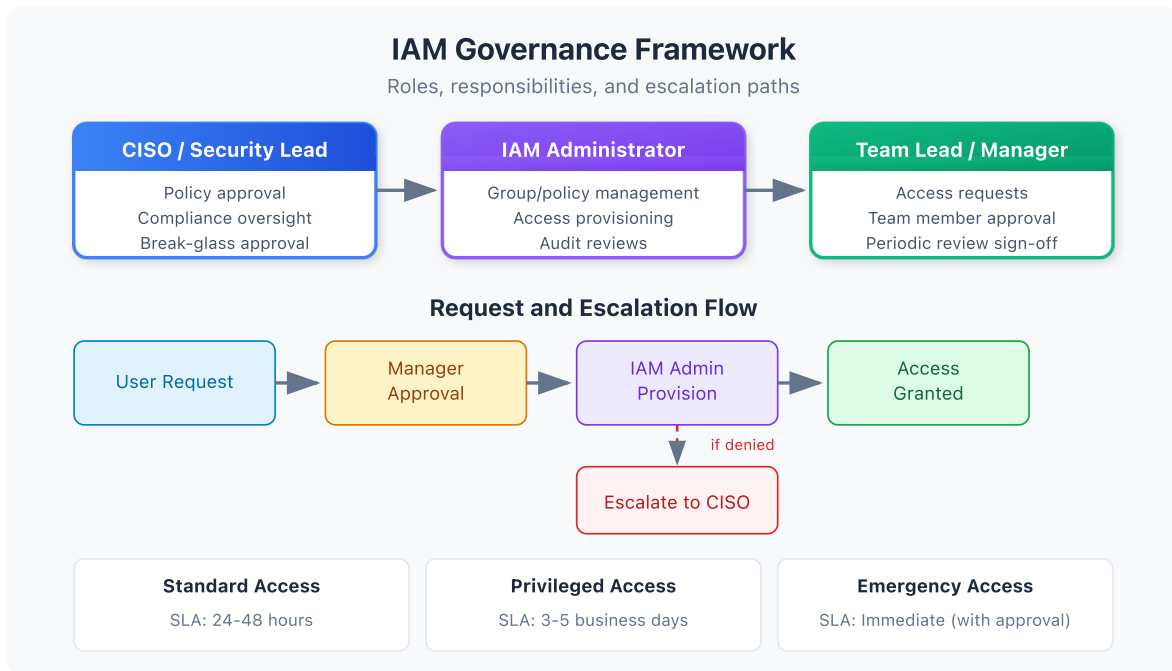
The Gen3 model provides several advantages over legacy Management Zones:

- **Scalability:** Policies and boundaries scale to thousands of entities
- **Flexibility:** Attribute-based rules adapt as your environment changes
- **Separation of Concerns:** Policies (actions) separate from boundaries (visibility)
- **Grail Integration:** Native support for Grail data lakehouse permissions
- **Bucket Partitioning:** Physical data isolation for compliance and team separation

Note: If you're migrating from Management Zones, see the **MZ2POL** series for migration guidance. This series assumes you're using Gen3 IAM.

2. The ABAC Model

Attribute-Based Access Control (ABAC) evaluates access requests based on attributes rather than static role assignments.



How Access is Evaluated

When a user attempts an action, Dynatrace evaluates:

1. **Identity**: Who is the user? What groups do they belong to?
2. **Policy**: Does any assigned policy grant this action?
3. **Boundary**: Is the target entity within their boundary conditions?
4. **Segment**: Does the data match their segment filters?

Access is granted only if ALL conditions are satisfied.

Policy Statement Structure

```
ALLOW <service>:<resource>:<action> [WHERE
<condition>]
```

Example:

```
ALLOW storage:logs:read WHERE storage:dt.security_context = "team-a"
```

Three Permission Domains

Domain	Controls	Examples
Environment	Smartscape entities (hosts, services)	View hosts, see topology
Storage	Grail data (logs, spans, metrics)	Read logs, query spans
Settings	Configuration objects	Edit alerting rules

3. Governance Roles and Responsibilities

Effective IAM governance requires clear ownership and accountability. Define these roles within your organization.

Core IAM Roles

Role	Responsibilities	Typical Team
Account Administrator	Manage account-level settings, billing, environments	Platform Team
IAM Administrator	Manage groups, policies, boundaries, user access	Security/IAM Team
Environment Administrator	Manage environment-specific settings and integrations	Platform Team
Security Auditor	Review access logs, compliance reporting	Security/Compliance
Group Owner	Manage membership of specific groups	Application Teams

Separation of Duties

Implement checks and balances:

Action	Primary	Approver
Create new policy	IAM Admin	Security Lead
Add user to admin group	IAM Admin	Account Admin
Modify boundaries	IAM Admin	Environment Owner
Create service account	IAM Admin	Application Owner

RACI Matrix for IAM Operations

Activity	Account Admin	IAM Admin	Env Admin	Auditor
Policy creation	A	R	C	I
Group management	I	R	C	I
Access reviews	A	R	C	R
Compliance reporting	I	C	I	R
Break-glass procedures	R	R	I	I

R=Responsible, A=Accountable, C=Consulted, I=Informed

4. Account vs Environment Permissions

Understanding the two-tier permission model is critical for proper governance.

Account Level

Account-level permissions apply across ALL environments:

Permission	Grants
<code>account-viewer</code>	View account settings, environments
<code>account-editor</code>	Modify account settings
<code>account-iam-admin</code>	Manage users, groups, account policies
<code>environment-creator</code>	Create new environments

Environment Level

Environment-level permissions scope to a single environment:

Permission	Grants
environment-viewer	Read-only access to environment data
environment-editor	Modify environment configurations
environment-admin	Full environment control
Custom policies	Fine-grained access per your design

Best Practice: Least Privilege

Account Level → Minimal, highly restricted

Environment Level → Scoped to team/function need

Example Structure:

User Type	Account Permission	Environment Permission
Platform Admin	account-iam-admin	environment-admin (all)
App Team Lead	account-viewer	environment-editor (their env)
Developer	None	Custom policy (read + limited write)
Auditor	account-viewer	environment-viewer (all)

5. Centralized vs Federated Models

Choose the governance model that fits your organization's structure.

Centralized Model

A single team controls all IAM operations.

Characteristics:

- Central IAM team manages all policies, groups, boundaries

- Consistent standards across the organization
- Single point of control and audit

Best For:

- Regulated industries (finance, healthcare)
- Organizations requiring strict compliance
- Smaller deployments (< 500 users)

Implementation:

Central IAM Team → All policy/group/boundary management
App Teams → Request access via tickets

Federated Model

Distributed ownership with central oversight.

Characteristics:

- Business units manage their own groups/boundaries
- Central team sets standards and audits
- Faster response to team needs

Best For:

- Large enterprises with autonomous teams
- Organizations with diverse tech stacks
- DevOps/platform engineering models

Implementation:

Central IAM Team → Policies, standards, audit
Group Owners → Group membership management
Environment Owners → Environment-specific boundaries

Hybrid Model

Most enterprises adopt a hybrid approach:

Area	Ownership
Account-level policies	Central IAM team
Environment policies	Central IAM team
Group membership	Delegated to group owners
Boundary definitions	Collaborative (central + env owners)

6. Documentation and Change Management

Governance requires documentation and controlled changes.

Required Documentation

Document	Content	Review Frequency
Access Policy	Who can access what	Annual
Group Catalog	All groups, owners, purposes	Quarterly
Policy Inventory	All policies with descriptions	Quarterly
Boundary Definitions	Boundary rules and rationale	Quarterly
Onboarding Guide	New user access process	As needed
Offboarding Checklist	Access revocation steps	As needed

Change Management Process

All IAM changes should follow a controlled process:

1. **Request:** Ticket with business justification
2. **Review:** IAM admin reviews against policy
3. **Approval:** Required approver signs off
4. **Implementation:** Change made with audit trail
5. **Verification:** Requestor confirms access works
6. **Documentation:** Update catalogs/inventories

GitOps for IAM

Consider version-controlling your IAM configuration:

```
# policies/team-a-readonly.yaml
name: team-a-readonly
description: Read-only access for Team A
statements:
  - effect: ALLOW
    service: storage
    resource: logs
    action: read
    conditions:
      - storage:dt.security_context = "team-a"
```

Benefits:

- Full audit trail via git history
- Peer review before changes
- Rollback capability
- Documentation as code

See **AUTOM-03: Monaco** for configuration-as-code implementation.

7. Governance Assessment

Use these queries to assess your current IAM governance posture.

```
// Count entities by security context – see how data is partitioned
fetch dt.entity.service
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missingContext = countIf(isNull(dt.security_context))
| fieldsAdd coveragePercent = round(100.0 * withContext / total, decimals: 2)
```

```
// List services without security context (governance gaps)
fetch dt.entity.service
| filter isNull(dt.security_context)
| fields entity.name, tags
| sort entity.name
| limit 50
```

```
// Review security context distribution
fetch dt.entity.service
| filter isNotNull(dt.security_context)
| summarize serviceCount = count(), by:{dt.security_context}
| sort serviceCount desc
| limit 20
```

```
// Check host security context coverage
fetch dt.entity.host
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missingContext = countIf(isNull(dt.security_context))
| fieldsAdd coveragePercent = round(100.0 * withContext / total, decimals: 2)
```

Interpreting Results

Metric	Target	Action if Below
Security context coverage	> 95%	Review tagging strategy
Orphaned entities	< 5%	Assign ownership
Context distribution	Balanced	Review boundary design

8. Next Steps

With governance foundations in place, proceed to implement the specific components:

Recommended Path

1. **IAMADM-02: SSO and Authentication** - Configure SSO before users and groups
2. **IAMADM-03: Group Architecture and Design** - Design your group hierarchy
3. **IAMADM-04: Policy Authoring and Management** - Create custom policies
4. **IAMADM-05: Boundary Design Patterns** - Implement data boundaries

Governance Checklist

Before moving on, ensure you have:

- [] Identified IAM roles and owners in your organization
 - [] Chosen centralized, federated, or hybrid model
 - [] Documented your access policy framework
 - [] Established a change management process
 - [] Assessed current security context coverage
-

Summary

In this notebook, you learned:

- The Gen3 IAM model and its components (policies, boundaries, segments)
 - How ABAC evaluates access requests
 - Core governance roles and their responsibilities
 - The difference between account and environment permissions
 - Centralized vs federated governance models
 - Documentation and change management requirements
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References

- [IAM Overview](#)
 - [Manage User Permissions with Policies](#)
 - [Account Management](#)
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